Password Security & Cracking Lab Report

1. Introduction

This lab demonstrates password hashing and cracking using Kali Linux tools.

Tools used: John the Ripper, Hashcat, OpenSSL.

Environment: Kali Linux Virtual Machine.

2. Methodology

Step 1: Tool Installation and Environment Setup

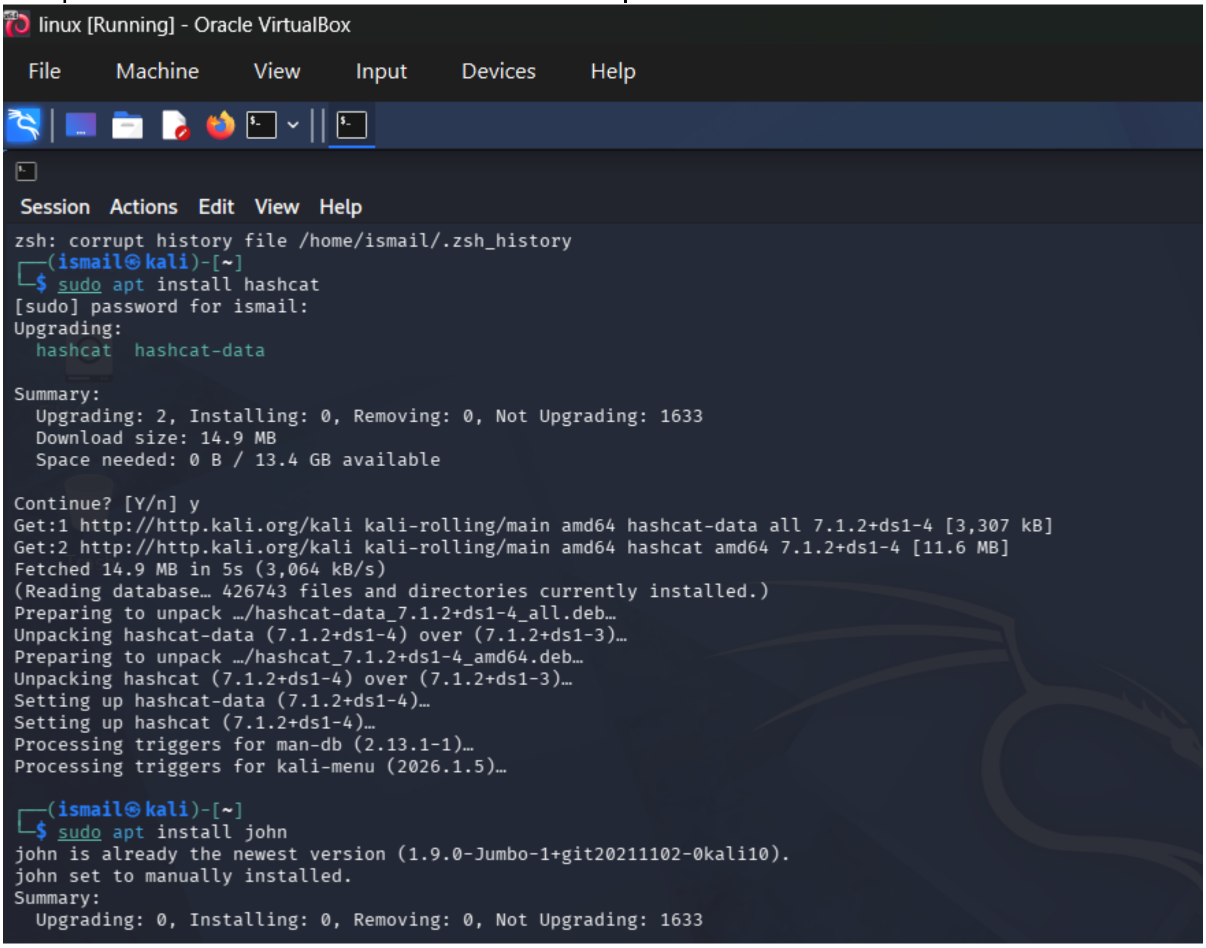

Figure 1: Installation of Hashcat and John the Ripper

Step 2: Wordlist Verification (rockyou.txt)

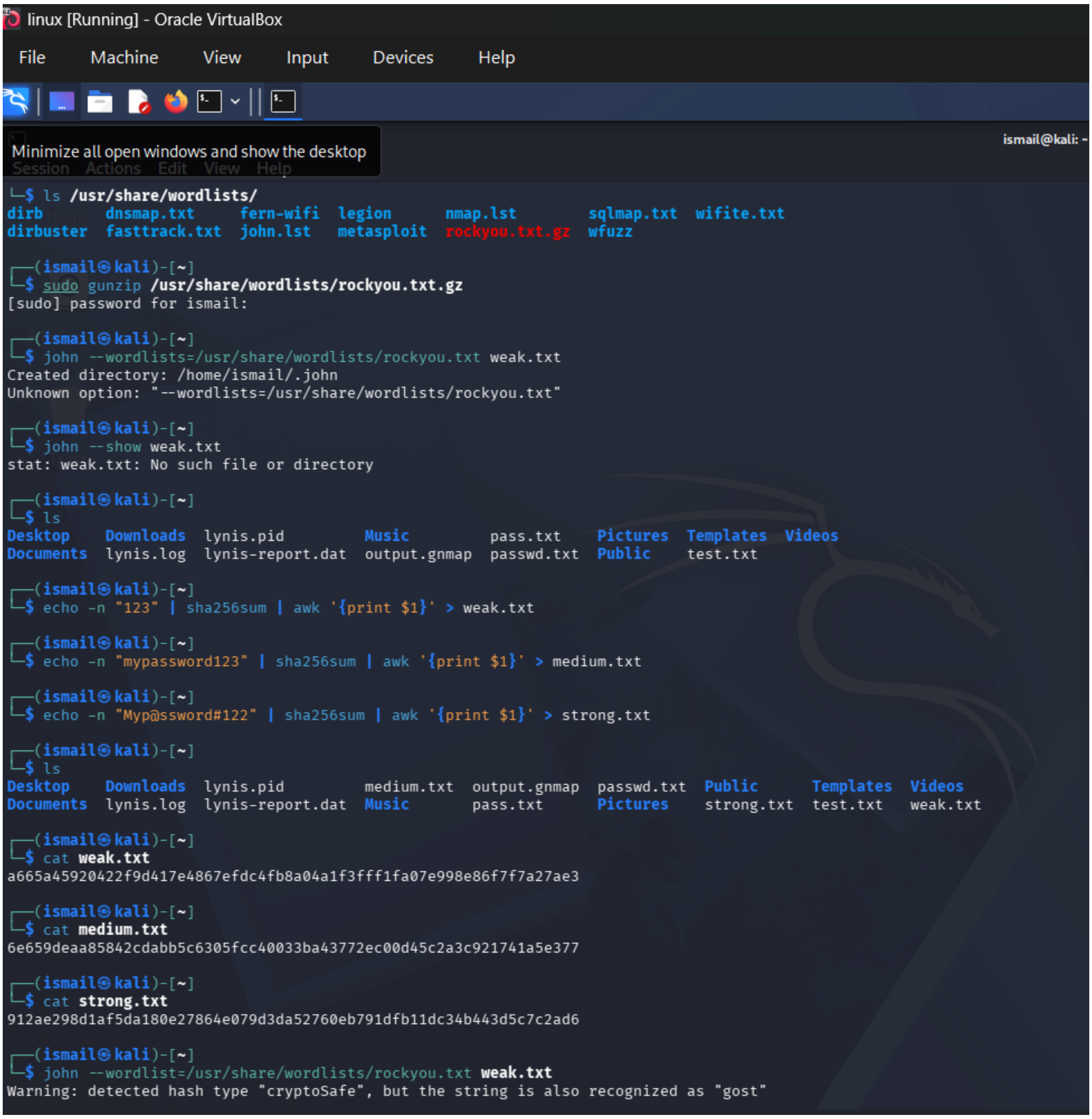

Figure 2: RockYou wordlist extraction and verification

Step 3: Password Hash Generation & John the Ripper

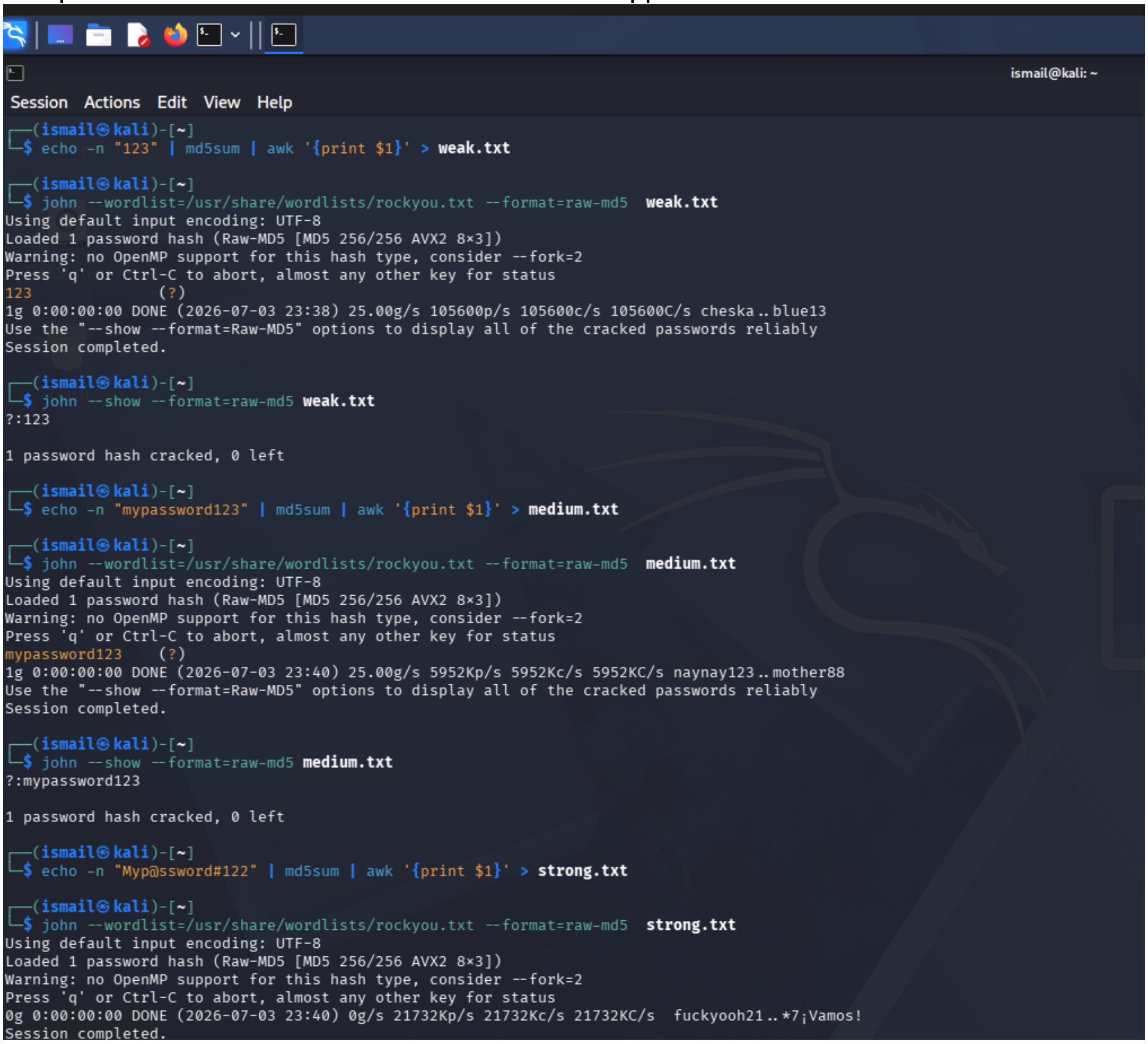

Figure 3: Password cracking using John the Ripper

Step 4: Hashcat Cracking (Weak & Medium)

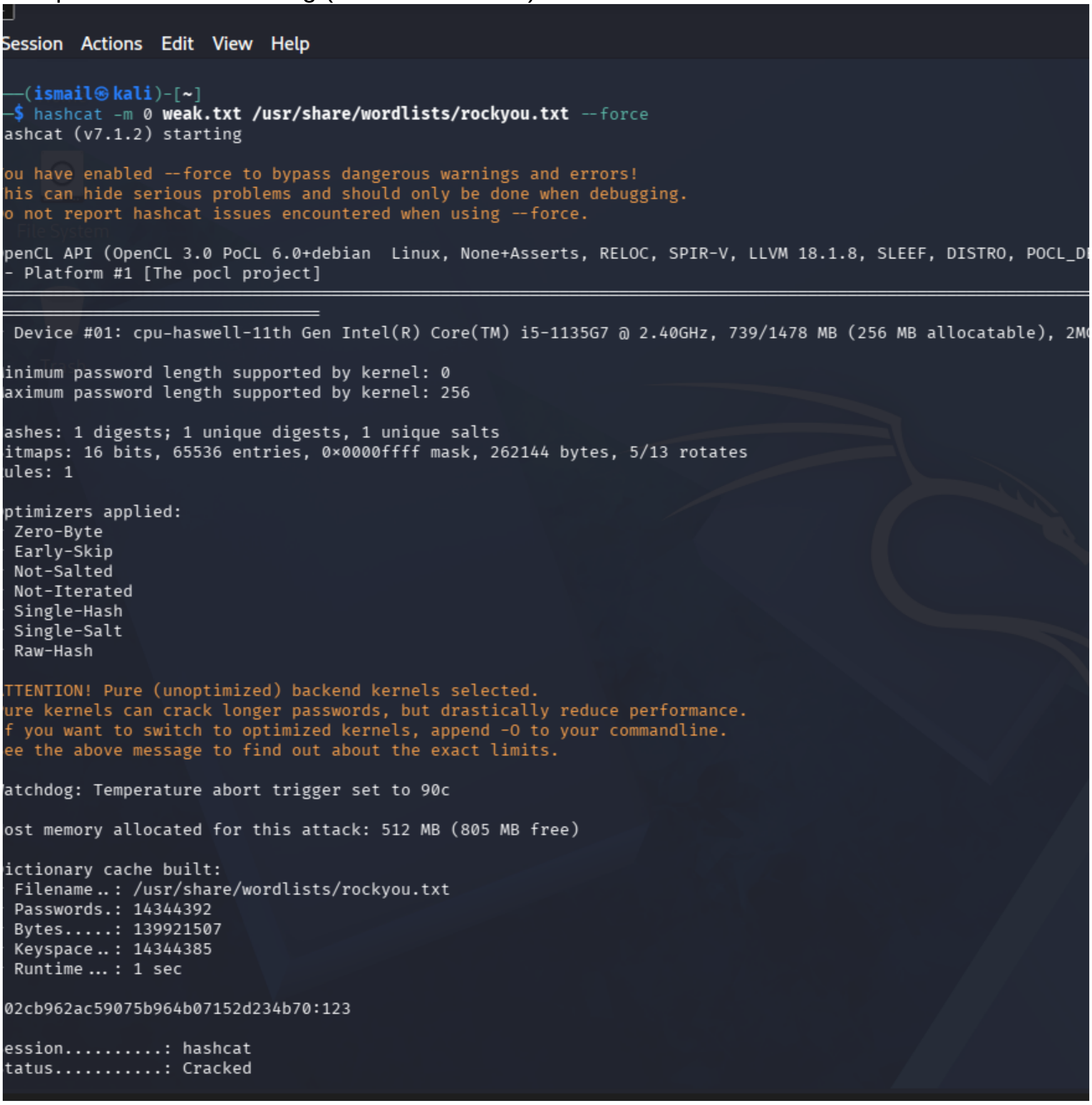

Figure 4: Hashcat cracking weak and medium passwords

Step 5: Hashcat Exhaustion (Strong Password)

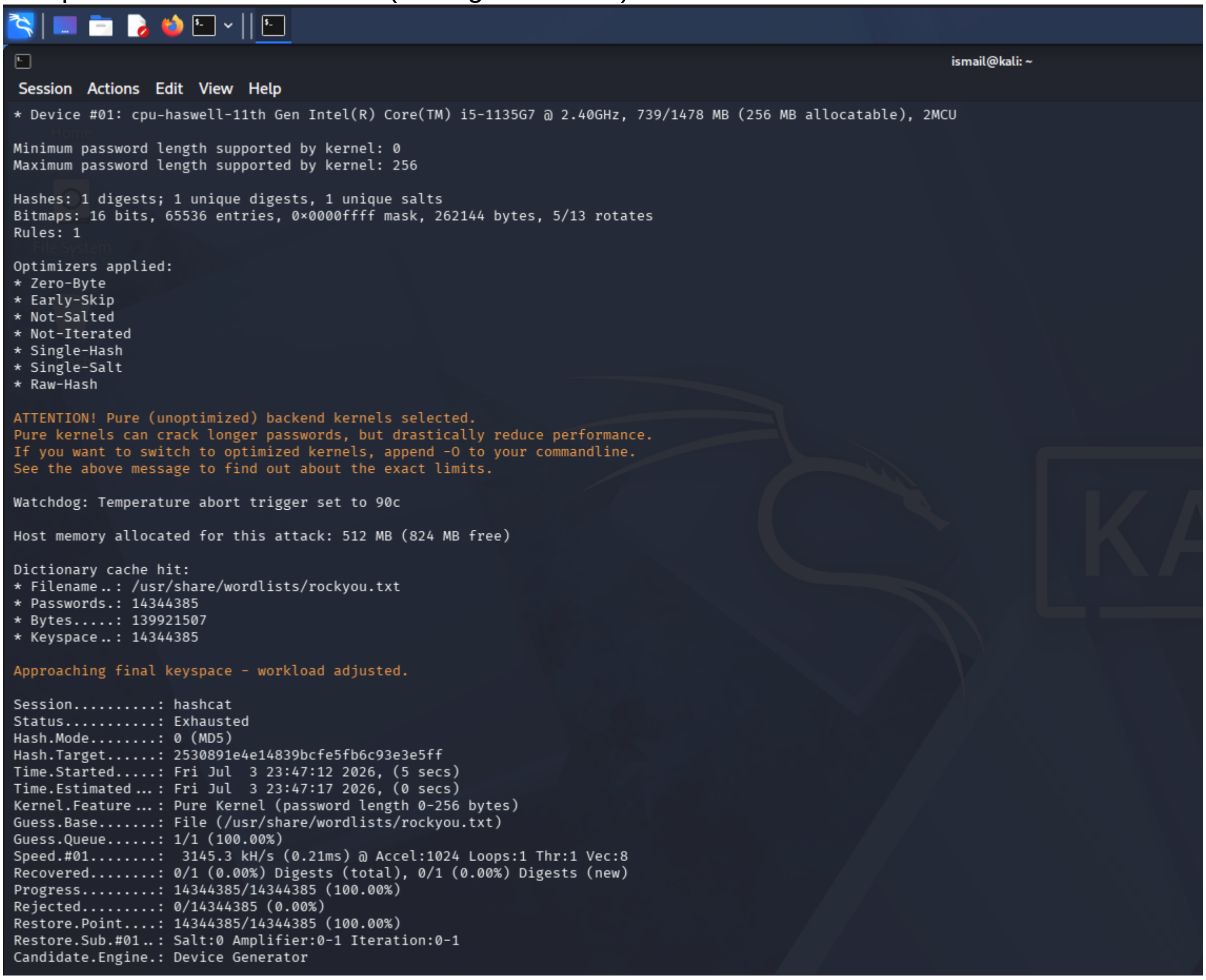

Figure 5: Strong password resisted cracking attempt

3. Observations

Weak passwords are cracked easily. Medium passwords show moderate resistance. Strong passwords resist cracking attempts.

MD5 is insecure and should not be used in production systems.

4. Recommendations

Use strong passwords with complexity and length.

Use modern hashing algorithms like bcrypt or Argon2.

Enable multi-factor authentication for better security.

5. Conclusion

This lab demonstrates the importance of password strength and secure hashing algorithms in cybersecurity.